

Test 1: Foundations of Physical Geography

Units 1–4 · pages 7–65 · 50 points · 50 minutes · Closed book

Name _____

Date _____

Score / 50

Read each question carefully. Circle one answer for multiple choice. Write on the lines provided; use the back of the page if you need more room. Point values are shown for each section.

Part A · Multiple Choice

12 questions × 2 points = 24 points

1. Which of the following correctly lists the components of Earth that physical geography studies?
 - a) Lithosphere, hydrosphere, biosphere, atmosphere, and interior
 - b) Crust, mantle, core, and atmosphere
 - c) Land, sea, and air only
 - d) Troposphere, stratosphere, mesosphere, and thermosphere
2. In a well-designed experiment, the one factor that the scientist deliberately manipulates is the
 - a) hypothesis
 - b) dependent variable
 - c) independent variable
 - d) experimental control
3. Which of the following is *not* a great circle?
 - a) The Equator
 - b) Any line of longitude
 - c) The circle of illumination
 - d) The Tropic of Capricorn
4. The observation that light from essentially every distant galaxy is redshifted supports the conclusion that
 - a) the universe is expanding
 - b) all galaxies formed at the same time
 - c) the Milky Way sits at the center of the universe
 - d) the universe is contracting
5. A galaxy that is egg-shaped, reddish to yellowish, and made up mostly of old stars with very little gas and dust is
 - a) an open cluster
 - b) a spiral galaxy
 - c) an elliptical galaxy
 - d) an irregular galaxy

6. Elements heavier than iron are created
- in the first few minutes after the Big Bang
 - inside white dwarfs
 - in the cores of ordinary main-sequence stars
 - during a supernova explosion
7. Which sequence lists the eons of geologic time from oldest to youngest?
- Hadean, Archean, Proterozoic, Phanerozoic
 - Archean, Hadean, Phanerozoic, Proterozoic
 - Phanerozoic, Proterozoic, Archean, Hadean
 - Proterozoic, Hadean, Archean, Phanerozoic
8. A geologist finds a sandstone layer lying beneath a shale layer in an undisturbed sequence. Which principle tells her the sandstone is older?
- Original horizontality
 - Superposition
 - Lateral continuity
 - Fossil succession
9. A mineral sample has a daughter-to-parent ratio of 1. How many half-lives have passed?
- None
 - One
 - Two
 - Ten
10. Which of the following is the *largest-scale* map?
- 1:1,000,000
 - 1:250,000
 - 1:100,000
 - 1:24,000
11. A cartographer is making a world map so that readers can compare how much cropland each country has in production. The best choice is
- a conformal projection such as Mercator
 - an equal-area projection such as Gall-Peters
 - an azimuthal equidistant projection
 - any projection, since all of them preserve area
12. On a topographic map drawn with a 20-foot contour interval, labeled index contours would appear every
- 20 feet
 - 40 feet
 - 100 feet
 - 200 feet

Part B · Matching

1 question × 8 points = 8 points

13. Write the letter of the correct definition next to each term.

- | | |
|----------------------------|--|
| _____ 1. Cartography | a) The study of layered sedimentary rocks |
| _____ 2. Cosmology | b) Using the geometry of triangles to determine a location |
| _____ 3. Stratigraphy | c) The study, practice, and interpretation of maps |
| _____ 4. Geocoding | d) Fossil assemblages are unique to the time in which they lived |
| _____ 5. Uniformitarianism | e) Reducing the complexity of the characteristics being mapped |
| _____ 6. Fossil succession | f) The study of the universe |
| _____ 7. Triangulation | g) Giving non-spatial data a geographic location |
| _____ 8. Generalization | h) “The present is the key to the past” |

Part C · Short Answer

4 questions × 3 points = 12 points

14. Explain the difference between *relative dating* and *absolute dating*, and name the scientific discovery that made absolute dating possible.

15. Why is the Equator the only parallel that is also a great circle?

16. Name the three geospatial technologies covered in Unit 1, and give one thing each is used for.

17. A star like our Sun and a star many times more massive both eventually run out of fuel. Describe how their final states differ.

Part D • Extended Response

1 question $\times$ 6 points = 6 points

18. Every flat map of the world is wrong in some way. Explain why this is unavoidable, name the three map properties a cartographer must choose among, and explain how a cartographer decides which one to preserve. Support your answer with at least two specific named projections.

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